Fatima Jinnah Women University



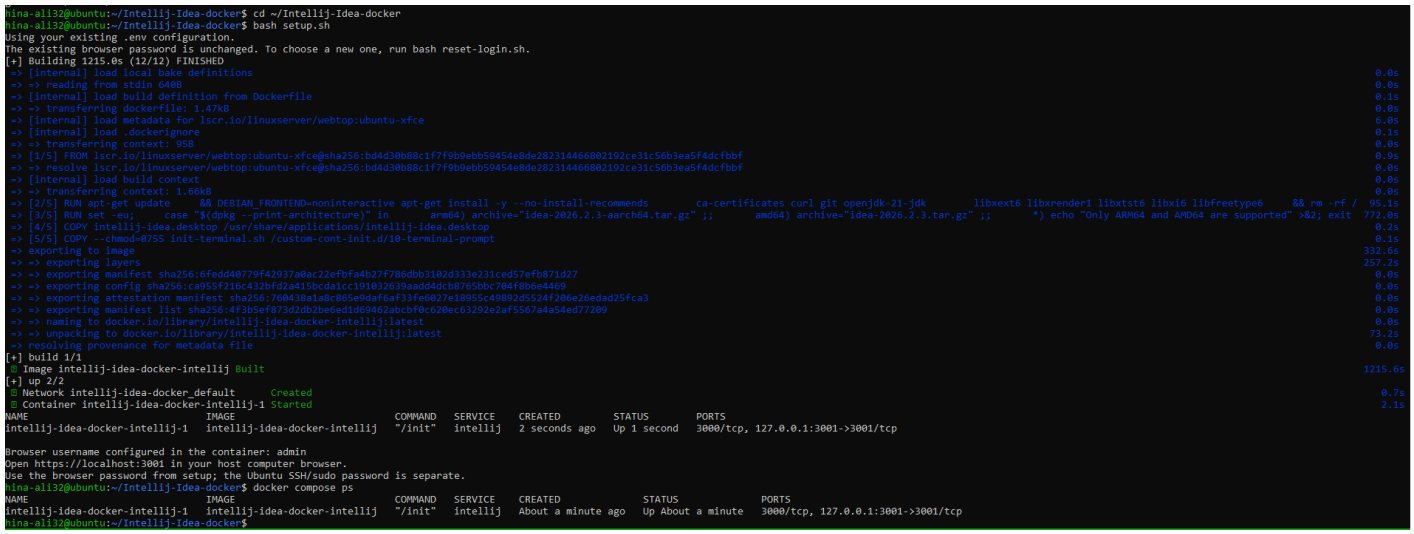
Department of Computer Science
DevOps Lab # 5

Name: Hina Ali
Registration no: 2023-BCS-040
Course code: CS-649
Section: BCS VII(B)
GitHub Username: Hina-Ali32

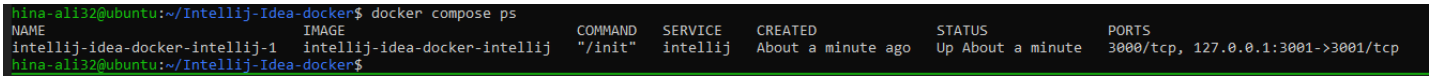
Task 1: Clone and Launch IntelliJ IDEA

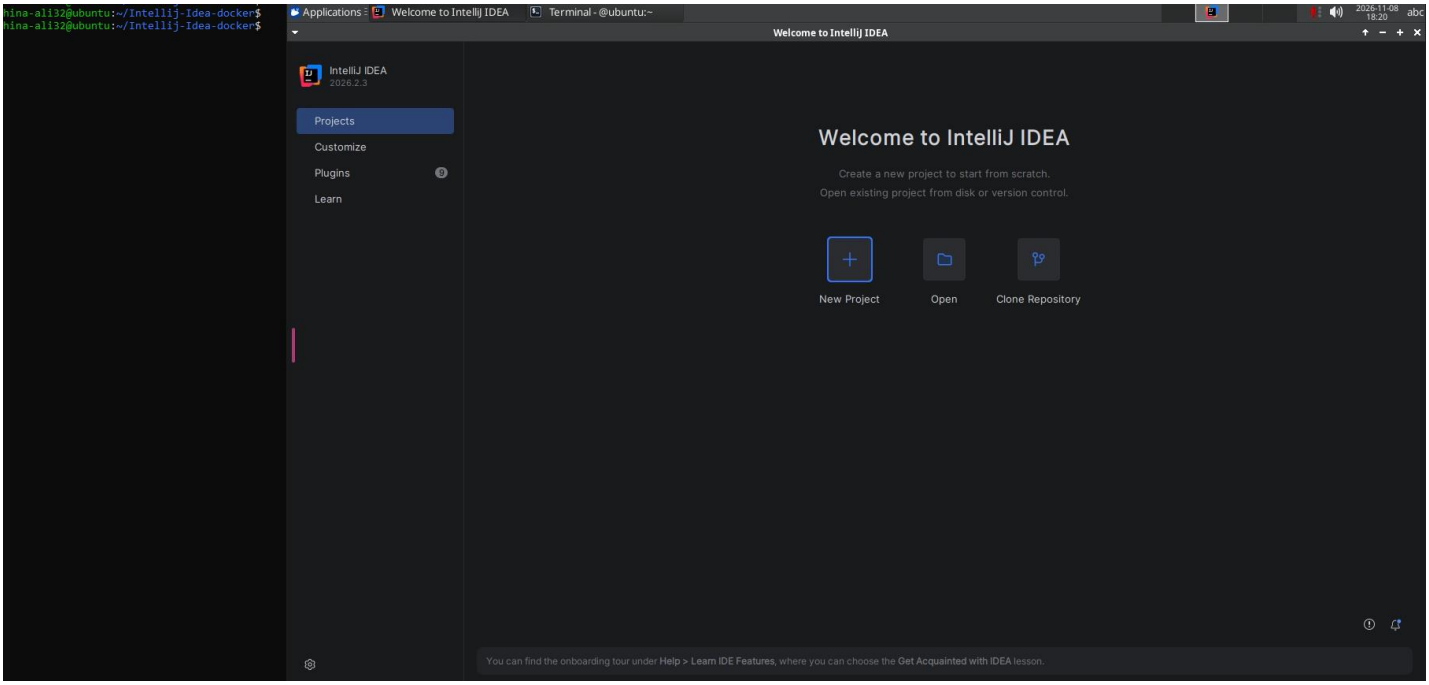
```
hina-ali32@ubuntu:~$ cd ~
hina-ali32@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Intellij-Idea-docker.git
Cloning into 'Intellij-Idea-docker'...
remote: Enumerating objects: 45, done.
remote: Counting objects: 100% (45/45), done.
remote: Compressing objects: 100% (41/41), done.
remote: Total 45 (delta 22), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (45/45), 19.02 KiB | 1.12 MiB/s, done.
Resolving deltas: 100% (22/22), done.
hina-ali32@ubuntu:~$ cd Intellij-Idea-docker
hina-ali32@ubuntu:~/Intellij-Idea-docker$
```

Shows the Intellij-Idea-docker repository cloned from GitHub on the Ubuntu host. The git clone command was run in the host terminal, and the output confirms the download completed. The repository contains the setup files for the IntelliJ desktop container.



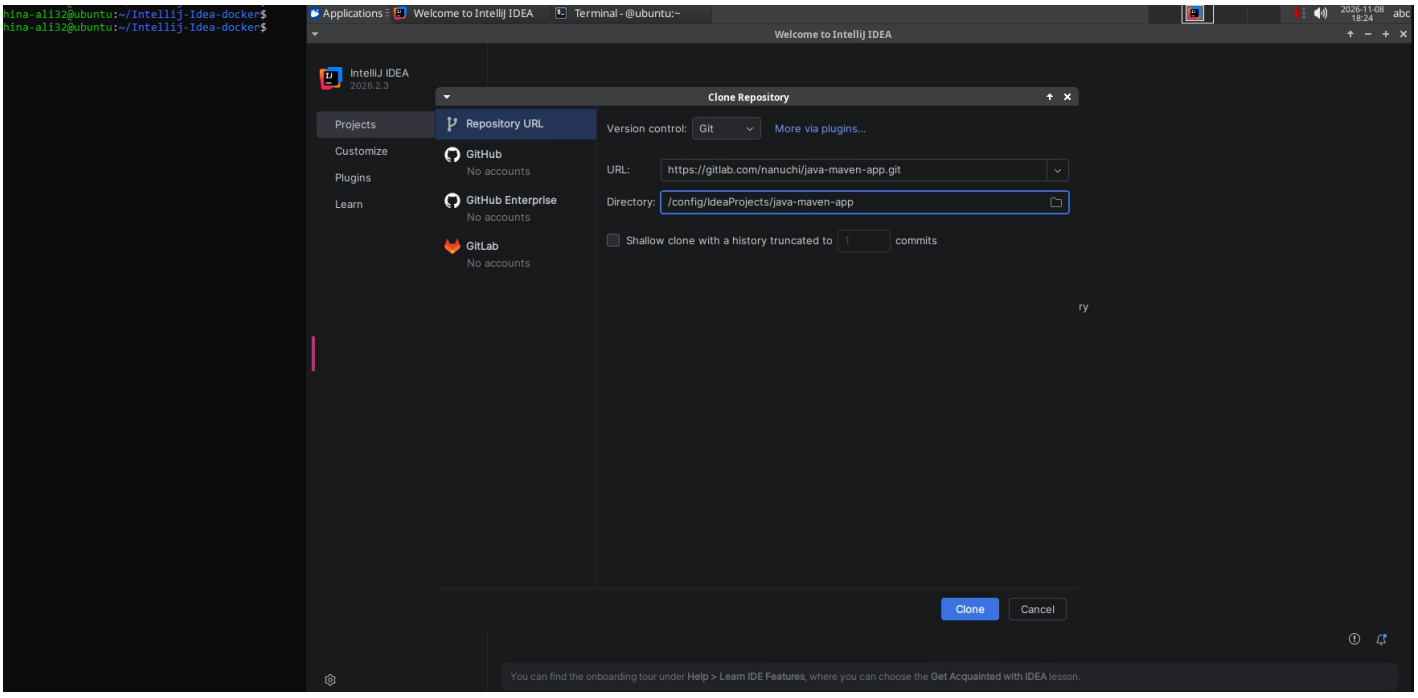
Shows the intellij service running with desktop port 3001. bash setup.sh built the image and started the container with Docker Compose. docker compose ps lists the service as Up with port 3001 mapped.





Shows the IntelliJ IDEA Welcome screen in the browser desktop beside the Ubuntu host terminal. The desktop was opened at https://localhost:3001 and IntelliJ was launched from the applications menu. The terminal prompt confirms the host identity.

Task 2: Build and Run the Maven Project



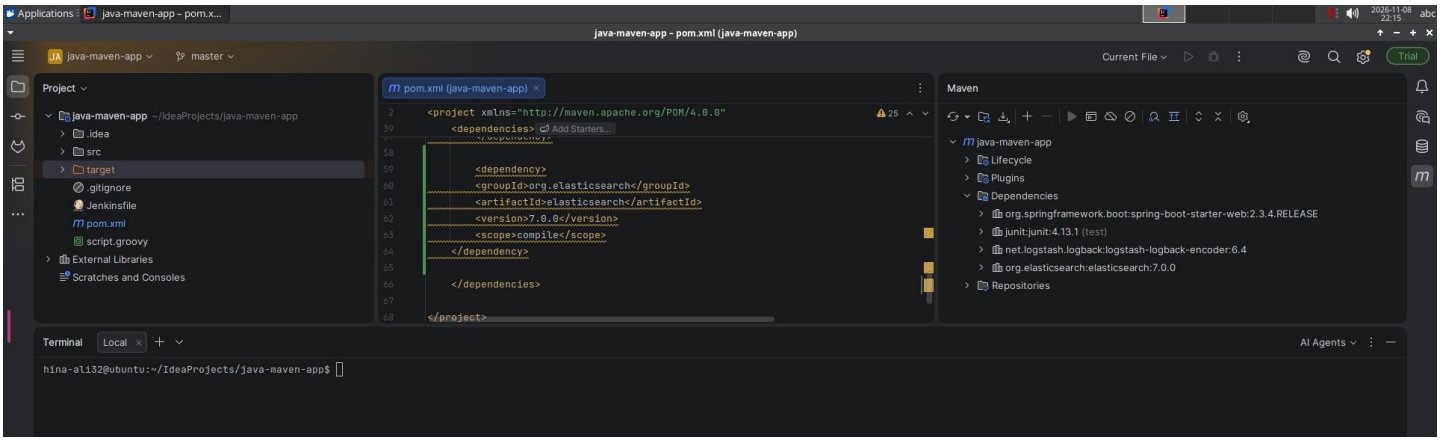
Shows the Clone Repository dialog with the java-maven-app URL and destination directory. The repository is cloned from GitLab into /config/IdeaProjects/java-maven-app, which is stored in the container's persistent volume.

```
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 21.0.12.1, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-38-generic", arch: "amd64", family: "unix"
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$
```

Shows the Maven version installed in IntelliJ's integrated terminal. Maven was installed with apt, and mvn --version confirms that the command is available.

```
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:java-maven-app >--------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  21.370 s
[INFO] Finished at: 2026-11-08T21:39:29+05:00
[INFO] ------------------------------------------------------------------------
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$
```

Shows mvn package completing with BUILD SUCCESS. The package phase compiles the source code and builds the project JAR in the target directory.



Shows the Elasticsearch 7.0.0 dependency added to pom.xml and the synchronized entry in the Maven tool window. The dependency was placed inside the existing <dependencies> section, and the Maven project was reloaded to download it.

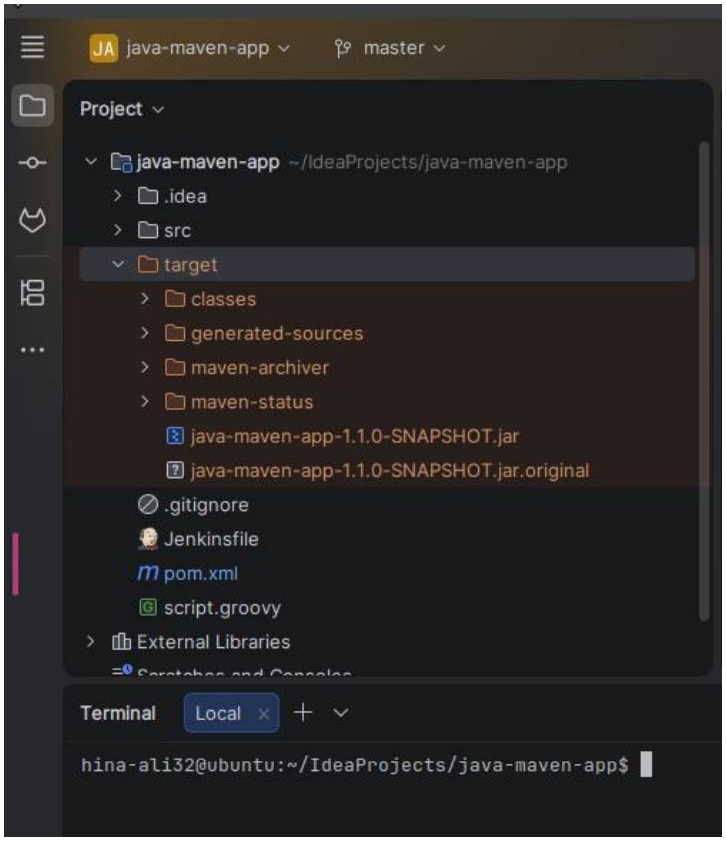
```
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:java-maven-app >---------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO] Building jar: /config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  20.832 s
[INFO] Finished at: 2026-11-08T22:17:29+05:00
[INFO] ------------------------------------------------------------------------
```

Shows mvn package completing again with BUILD SUCCESS after the dependency change. This confirms that the project still builds with Elasticsearch 7.0.0 included.

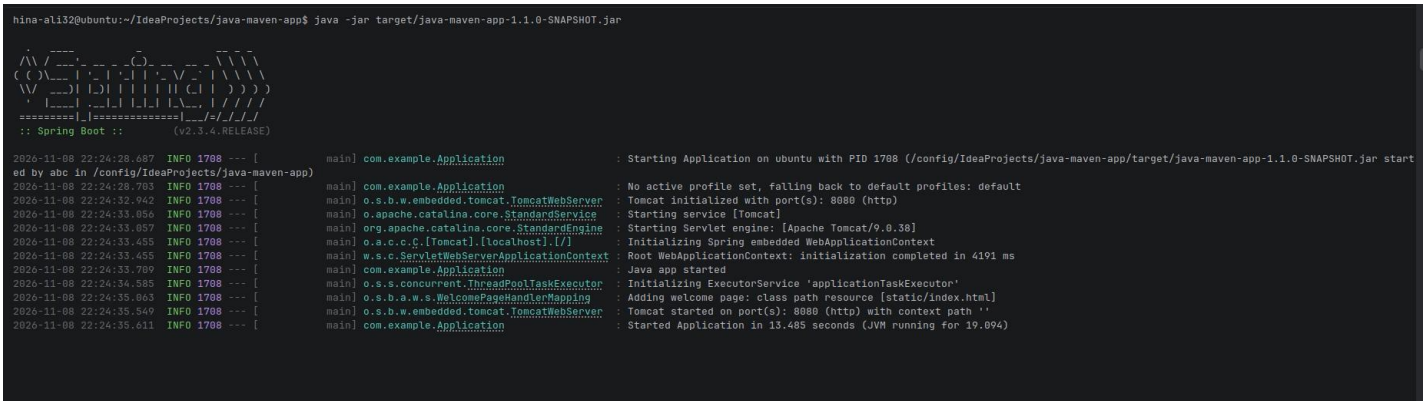
```
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:java-maven-app >--------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.667 s
[INFO] Finished at: 2026-11-08T22:19:27+05:00
[INFO] ------------------------------------------------------------------------

hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$
```

Shows mvn test running Maven's test phase and finishing with BUILD SUCCESS. The sample project has no test source files, so the output may report that no tests were run, which is the expected result for this project.



Shows the target directory containing java-maven-app-1.1.0-SNAPSHOT.jar. The file name comes from the artifactId and version values in pom.xml.



Shows the JAR command and the Spring Boot application starting on port 8080. The output reports Tomcat starting on port 8080 and the line Started Application, which confirms a successful startup.

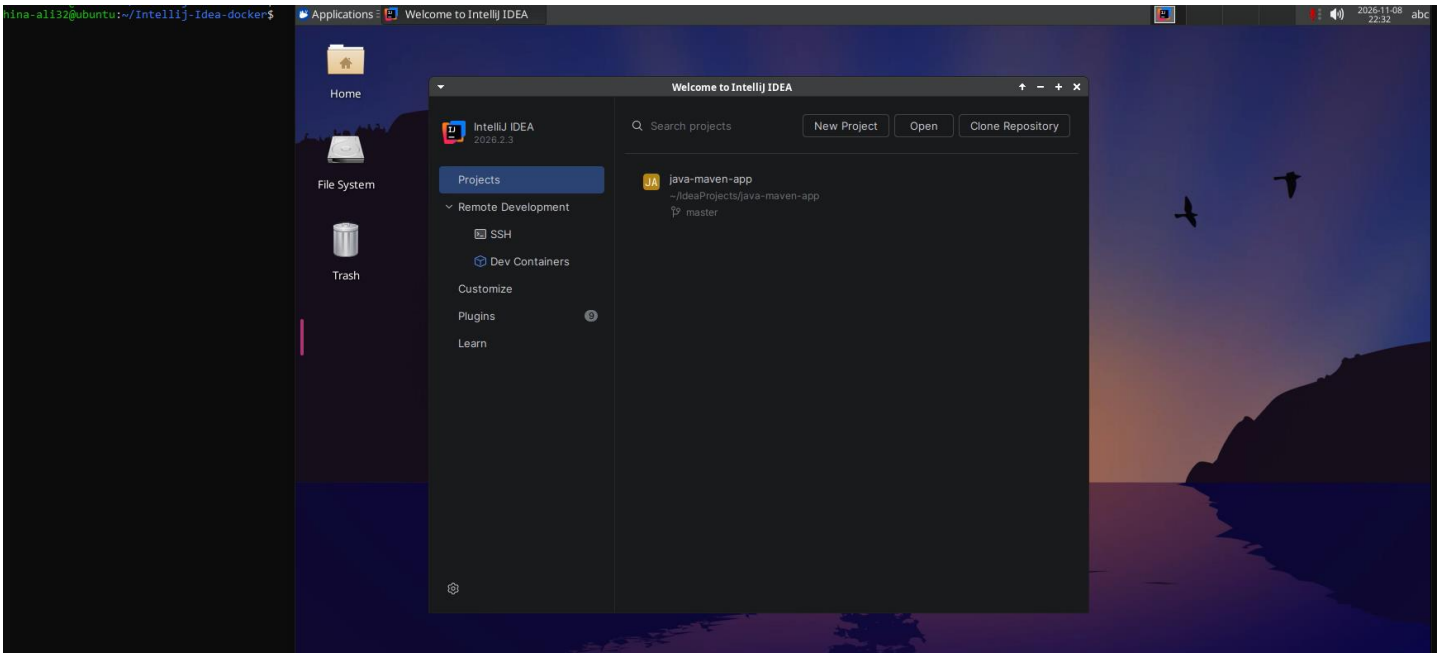
```
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$ java -jar target/java-maven-app-1.1.0-SNAPSHOT.jar

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::        (v2.3.4.RELEASE)

2026-11-08 22:24:28.687  INFO 1708 --- [           main] com.example.Application                  : Starting Application on ubuntu with PID 1708 (/config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar start
ed by abc in /config/IdeaProjects/java-maven-app)
2026-11-08 22:24:28.703  INFO 1708 --- [           main] com.example.Application                  : No active profile set, falling back to default profiles: default
2026-11-08 22:24:32.942  INFO 1708 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port(s): 8080 (http)
2026-11-08 22:24:33.056  INFO 1708 --- [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-11-08 22:24:33.057  INFO 1708 --- [           main] org.apache.catalina.core.StandardEngine  : Starting Servlet engine: [Apache Tomcat/9.0.38]
2026-11-08 22:24:33.455  INFO 1708 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
2026-11-08 22:24:33.455  INFO 1708 --- [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 4191 ms
2026-11-08 22:24:33.709  INFO 1708 --- [           main] com.example.Application                  : Java app started
2026-11-08 22:24:34.585  INFO 1708 --- [           main] o.s.s.concurrent.ThreadPoolTaskExecutor  : Initializing ExecutorService 'applicationTaskExecutor'
2026-11-08 22:24:35.063  INFO 1708 --- [           main] o.s.b.a.w.s.WelcomePageHandlerMapping    : Adding welcome page: class path resource [static/index.html]
2026-11-08 22:24:35.549  INFO 1708 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port(s): 8080 (http) with context path ''
2026-11-08 22:24:35.611  INFO 1708 --- [           main] com.example.Application                  : Started Application in 13.485 seconds (JVM running for 19.094)
^C2026-11-08 22:28:44.780  INFO 1708 --- [extShutdownHook] o.s.s.concurrent.ThreadPoolTaskExecutor  : Shutting down ExecutorService 'applicationTaskExecutor'

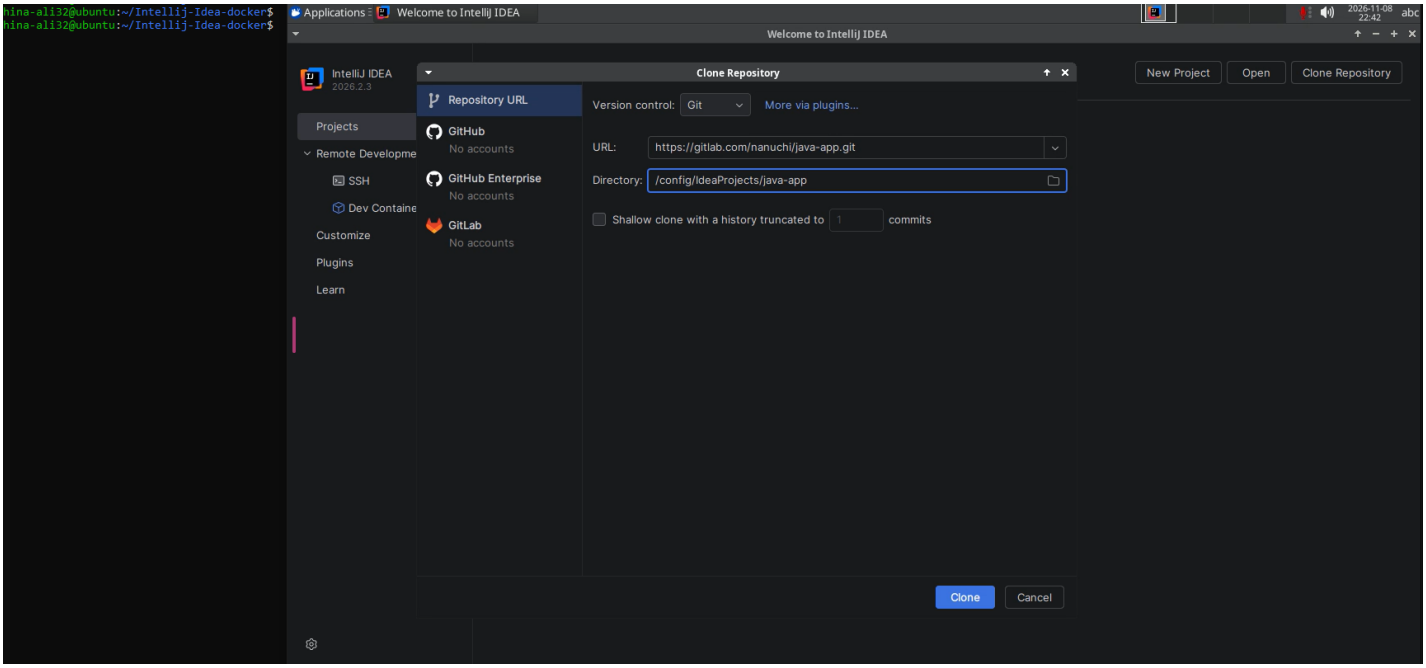
hina-ali32@ubuntu:~/IdeaProjects/java-maven-app$
```

Shows the application stopped with Ctrl+C and the shell prompt returned. The terminal is free again and port 8080 is released.

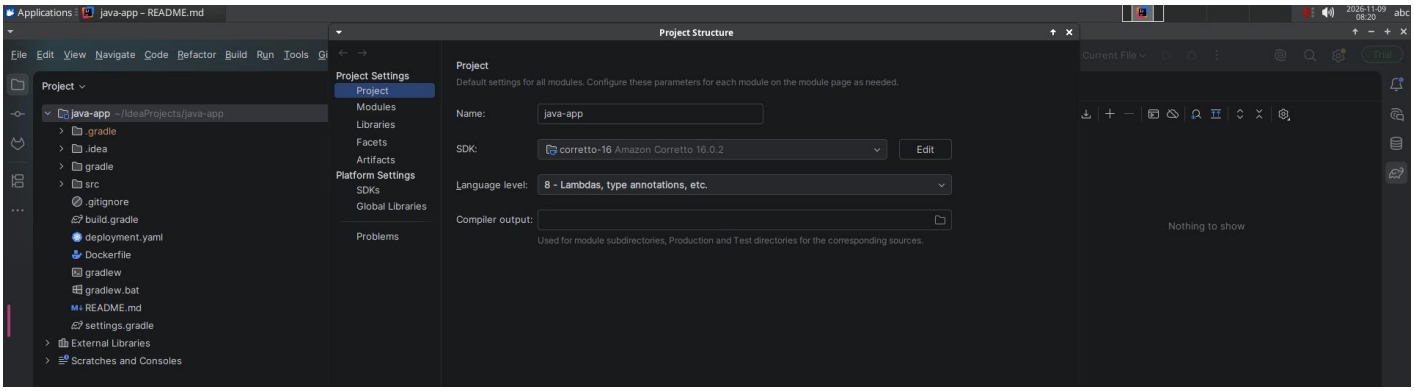


Shows the IntelliJ Welcome screen after closing the Maven project, beside the Ubuntu host terminal. The host prompt provides the required identity evidence.

Task 3: Build and Run the Gradle Project



Shows the Clone Repository dialog with the java-app URL and destination directory. The repository is cloned from GitLab into /config/IdeaProjects/java-app.



```
hina-ali32@ubuntu:~/IdeaProjects/java-app$ ./gradlew --version

------------------------------------------------------------
Gradle 7.0
------------------------------------------------------------

Build time:   2021-04-09 22:27:31 UTC
Revision:     d5661e3f0e07a8caff705f1badf79fb5df8022c4

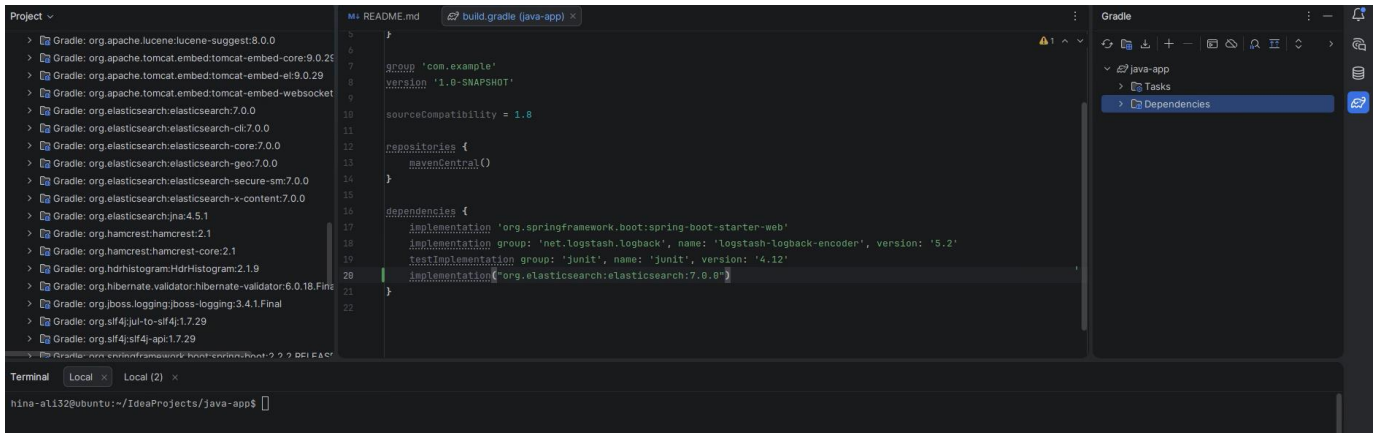
Kotlin:       1.4.31
Groovy:       3.0.7
Ant:          Apache Ant(TM) version 1.10.9 compiled on September 27 2020
JVM:          16.0.2 (Amazon.com Inc. 16.0.2+7)
OS:           Linux 7.0.0-38-generic amd64
```

```
hina-ali32@ubuntu:~/IdeaProjects/java-app$ ./gradlew build
Starting a Gradle Daemon, 1 stopped Daemon could not be reused, use --status for details

BUILD SUCCESSFUL in 6m 18s
3 actionable tasks: 3 executed

hina-ali32@ubuntu:~/IdeaProjects/java-app$
```

Shows ./gradlew build completing with BUILD SUCCESSFUL. The Gradle Wrapper ran on JDK 16, because the supplied Gradle 7.0 cannot run on Java 21.



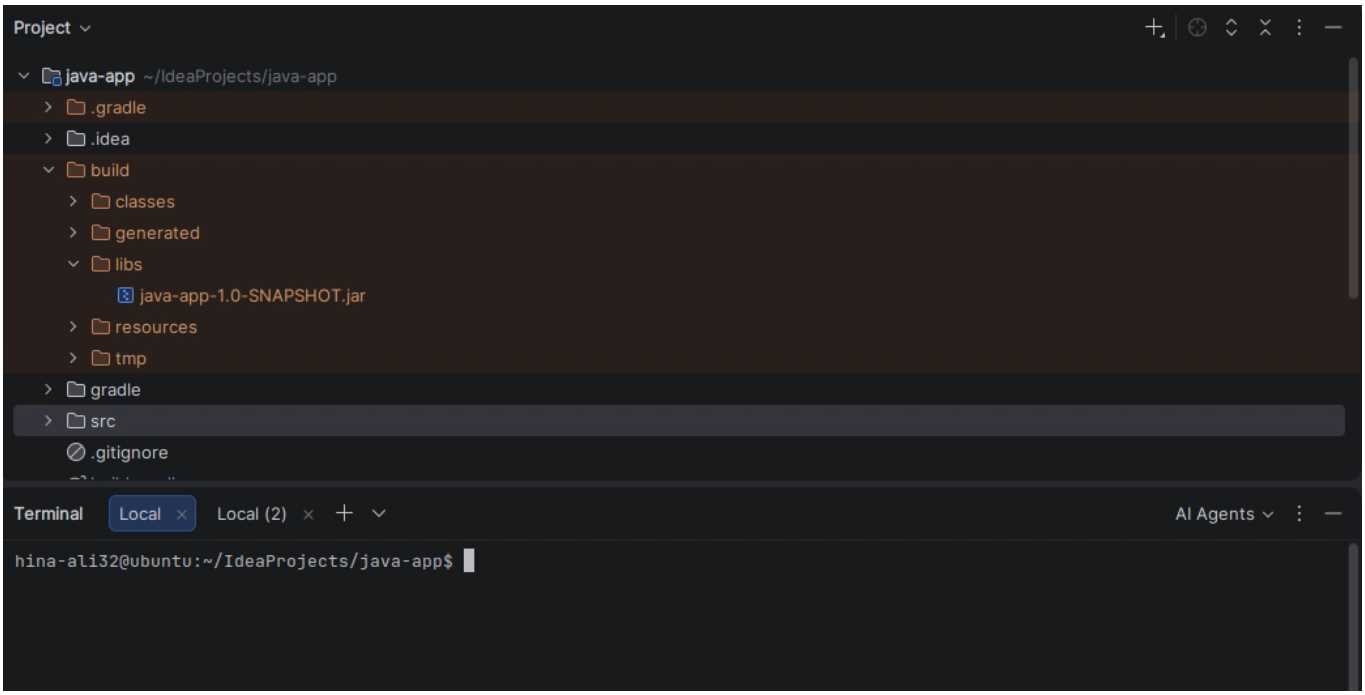
Shows the Elasticsearch 7.0.0 dependency added to build.gradle and the synchronized library under External Libraries. The implementation line was added inside the existing dependencies block.

```
hina-ali32@ubuntu:~/IdeaProjects/java-app$ ./gradlew build

BUILD SUCCESSFUL in 2m 15s
3 actionable tasks: 2 executed, 1 up-to-date

hina-ali32@ubuntu:~/IdeaProjects/java-app$
```

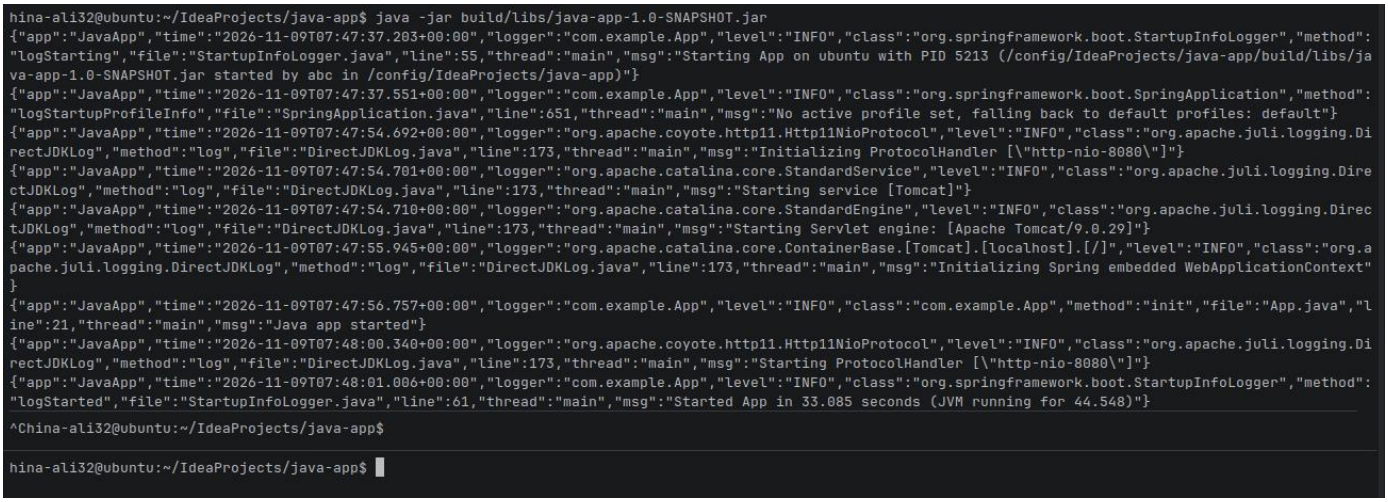
Shows the rebuild completing with BUILD SUCCESSFUL after adding the dependency. Gradle resolved the new library and rebuilt the project.



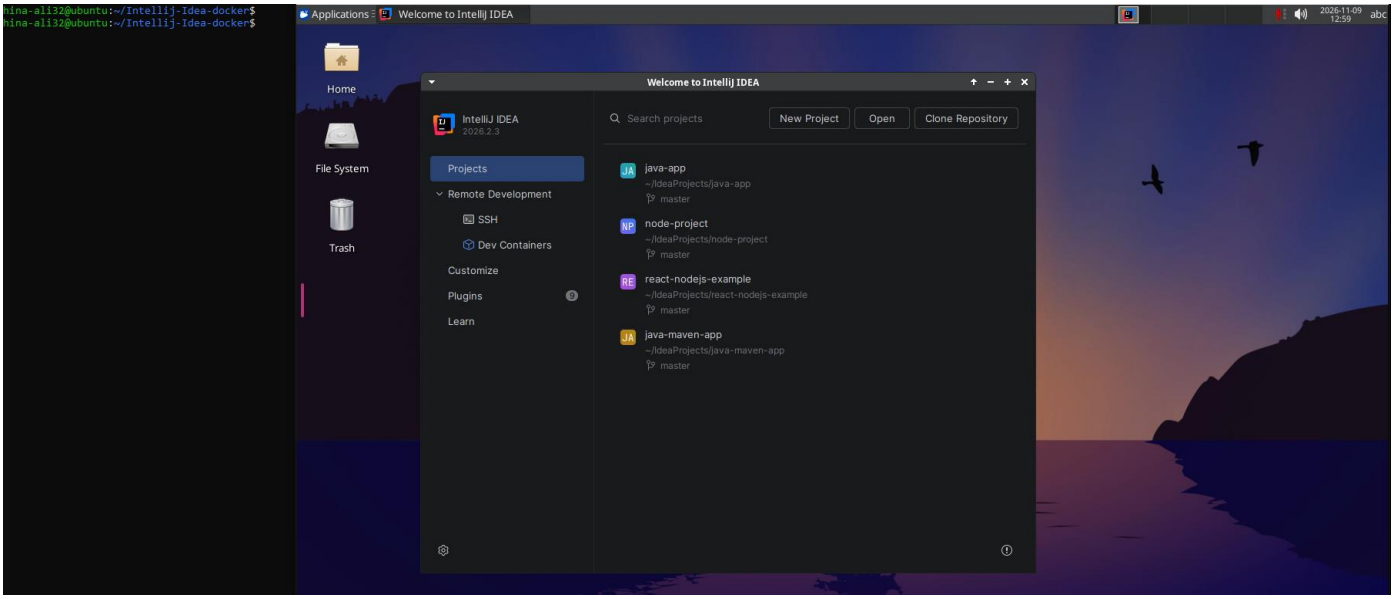
Shows the build/libs directory containing java-app-1.0-SNAPSHOT.jar. This is the executable JAR generated by the Gradle build.



Shows the JAR command and the application starting successfully. The application was launched with java -jar from the project root.

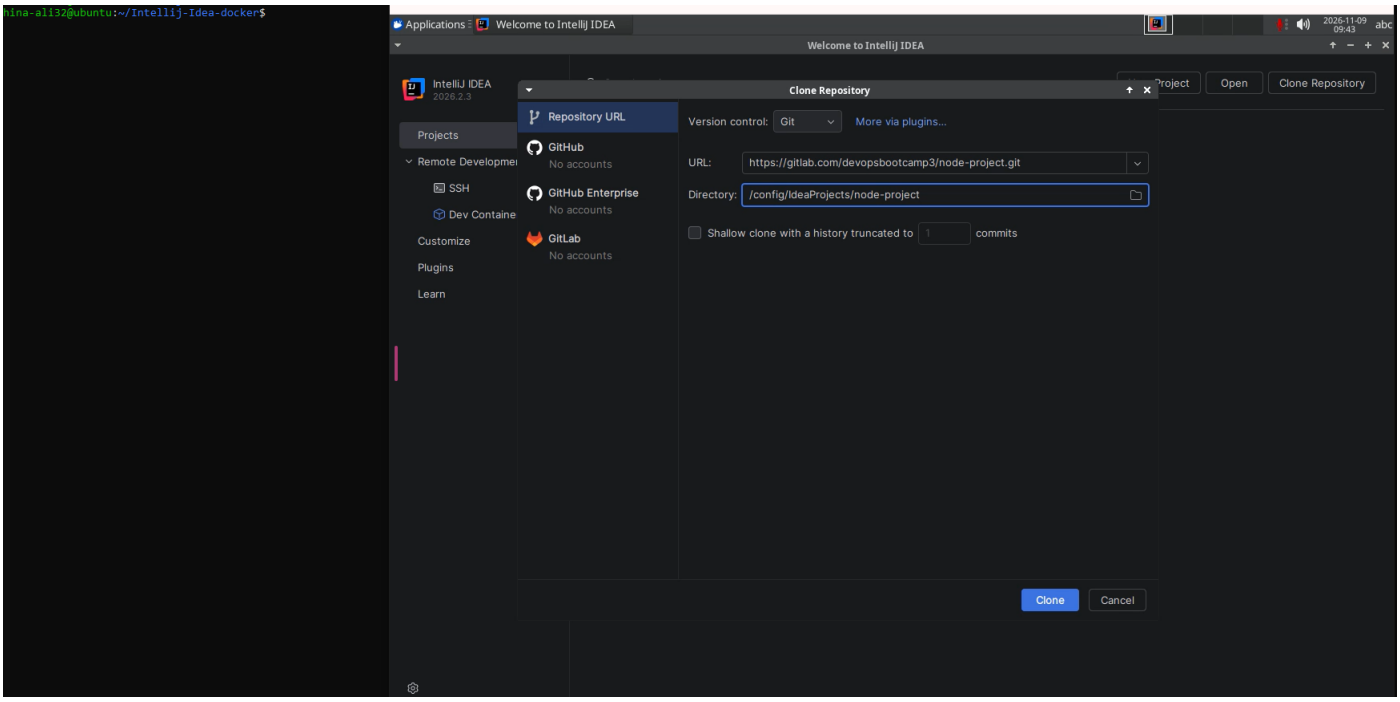


Shows the application stopped with Ctrl+C and the prompt returned. Stopping it also releases port 8080 for other applications.



Shows the IntelliJ Welcome screen after closing the Gradle project, beside the Ubuntu host terminal**.**

Task 4: Install and Run the Node.js Project



Shows the Clone Repository dialog with the node-project URL and destination directory. The

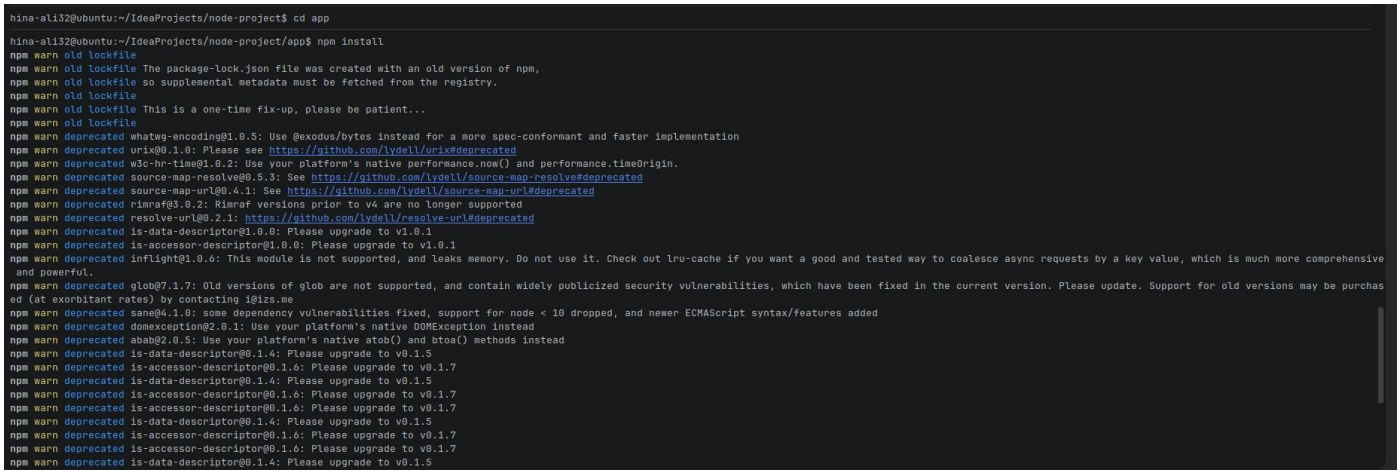
project is cloned into /config/IdeaProjects/node-project, and its application files are in the app subdirectory.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project$ node --version
v22.23.4

hina-ali32@ubuntu:~/IdeaProjects/node-project$ npm --version
10.9.9

hina-ali32@ubuntu:~/IdeaProjects/node-project$
```

Shows the installed Node.js and npm versions in IntelliJ's terminal. Node.js v22.23.4 and npm 10.9.9 are reported by node --version and npm --version.




```
hina-ali32@ubuntu:~/IdeaProjects/node-project/app$ npm install

up to date, audited 555 packages in 17s

24 packages are looking for funding
  run `npm fund` for details

54 vulnerabilities (10 low, 20 moderate, 20 high, 4 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.

hina-ali32@ubuntu:~/IdeaProjects/node-project/app$
```

Shows npm install completing in the app directory. The dependencies listed in package.json were downloaded into node_modules.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project/app$ npm start

> bootcamp-node-project@1.0.0 start
> node server.js

node:events:497
      throw er; // Unhandled 'error' event
      ^

Error: listen EADDRINUSE: address already in use :::3000
    at Server.setupListenHandle [as _listen2] (node:net:1941:16)
    at listenInCluster (node:net:1998:12)
    at Server.listen (node:net:2103:7)
    at Function.listen (/config/IdeaProjects/node-project/app/node_modules/express/lib/application.js:618:24)
    at Object.<anonymous> (/config/IdeaProjects/node-project/app/server.js:23:5)
    at Module._compile (node:internal/modules/cjs/loader:1781:14)
    at Object..js (node:internal/modules/cjs/loader:1913:10)
    at Module.load (node:internal/modules/cjs/loader:1505:32)
    at Function._load (node:internal/modules/cjs/loader:1309:12)
    at wrapModuleLoad (node:internal/modules/cjs/loader:254:19)
Emitted 'error' event on Server instance at:
    at emitErrorNT (node:net:1977:8)
    at process.processTicksAndRejections (node:internal/process/task_queues:89:21) {
  code: 'EADDRINUSE',
  errno: -98,
  syscall: 'listen',
  address: '::',
  port: 3000
}

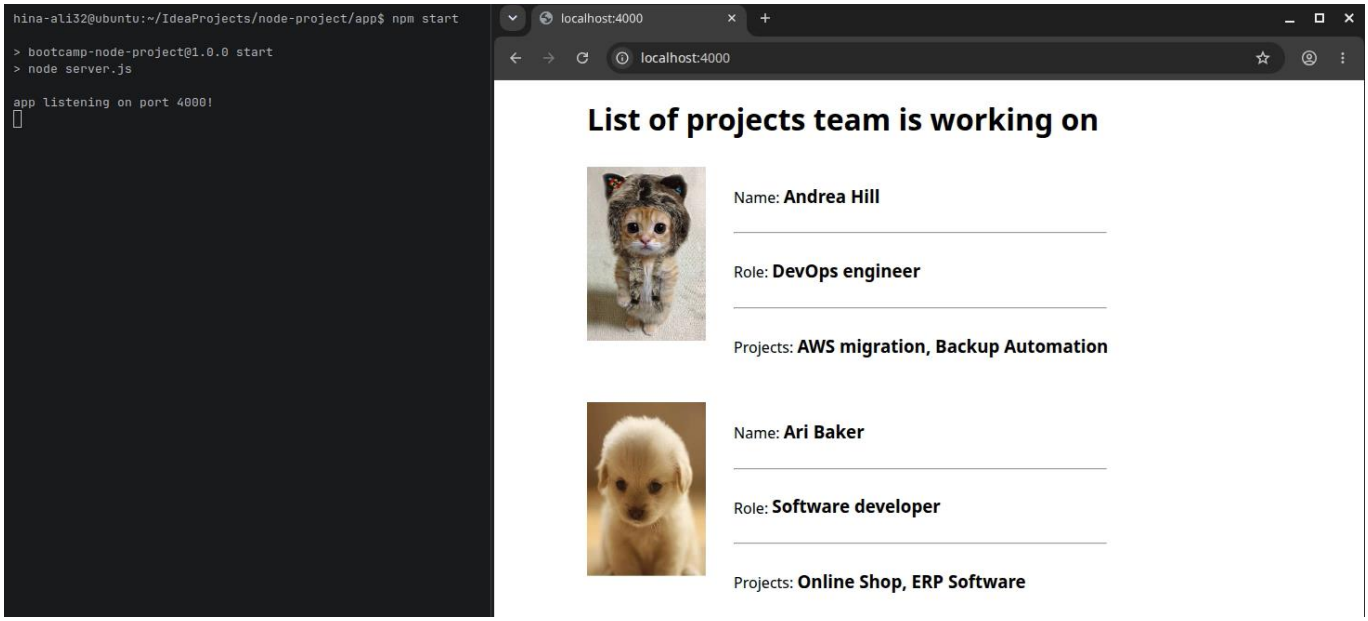
Node.js v22.23.4
```

Shows npm start running with the message app listening on port 4000!. Port 3000 caused an EADDRINUSE error because the desktop already uses it, so the listener in server.js was changed to port 4000.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project/app$ npm start

> bootcamp-node-project@1.0.0 start
> node server.js

app listening on port 4000!
```



Shows the Node.js project page at http://localhost:4000 beside IntelliJ's terminal. The page and its profile images load inside the container's browser, which confirms the application is serving requests.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project/app$ npm test

> bootcamp-node-project@1.0.0 test
> jest

Browserslist: caniuse-lite is outdated. Please run:
npx browserslist@latest --update-db

Why you should do it regularly:
https://github.com/browserslist/browserslist#browsers-data-updating
 PASS  ./server.test.js
  ✓ main index.html file exists (2 ms)

Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   0 total
Time:        6.054 s
Ran all test suites.

hina-ali32@ubuntu:~/IdeaProjects/node-project/app$
```

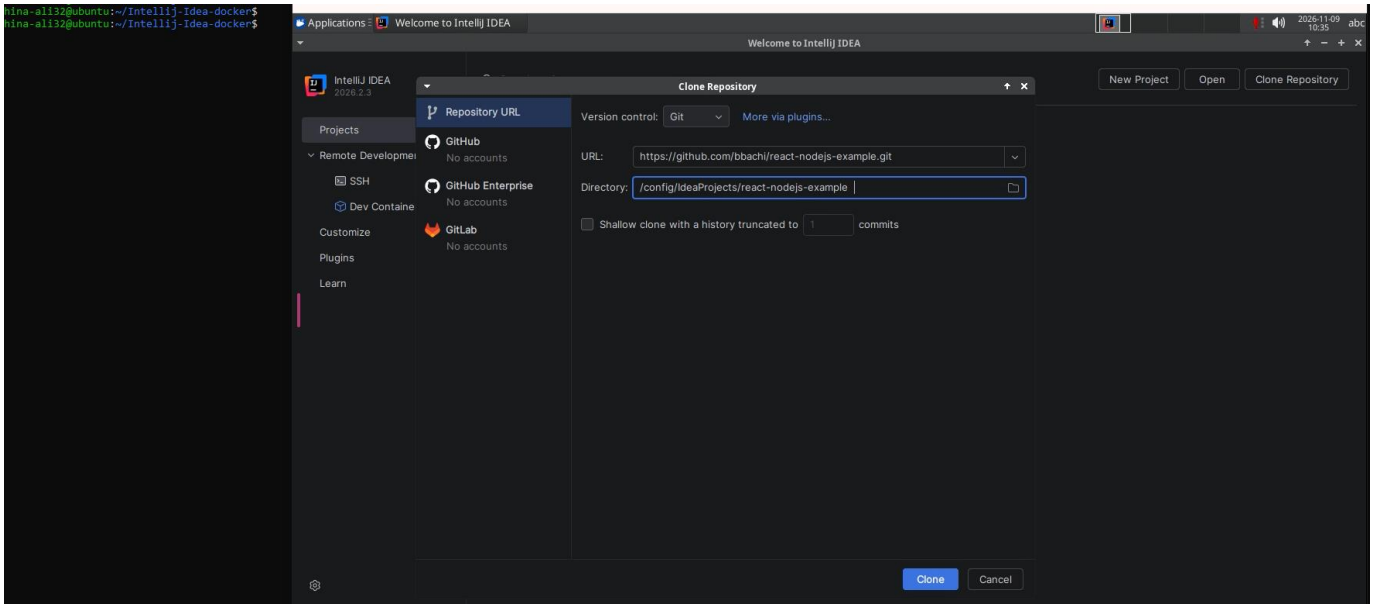
Shows npm test finishing with a passing result. The Jest test checks that index.html exists.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project/app$ npm pack
npm notice
npm notice 📦  bootcamp-node-project@1.0.0
npm notice Tarball Contents
npm notice 60.5kB images/profile-andrea.jpg
npm notice 17.5kB images/profile-ari.jpeg
npm notice 1.2kB index.html
npm notice 283B package.json
npm notice 723B server.js
npm notice 205B server.test.js
npm notice Tarball Details
npm notice name: bootcamp-node-project
npm notice version: 1.0.0
npm notice filename: bootcamp-node-project-1.0.0.tgz
npm notice package size: 79.4 kB
npm notice unpacked size: 80.4 kB
npm notice shasum: 3cd3f8718e68ba6305e99fc54673830bc468b54b
npm notice integrity: sha512-PLEYHDvH6C6lG[...]RHylrii4ybZEg==
npm notice total files: 6
npm notice
bootcamp-node-project-1.0.0.tgz

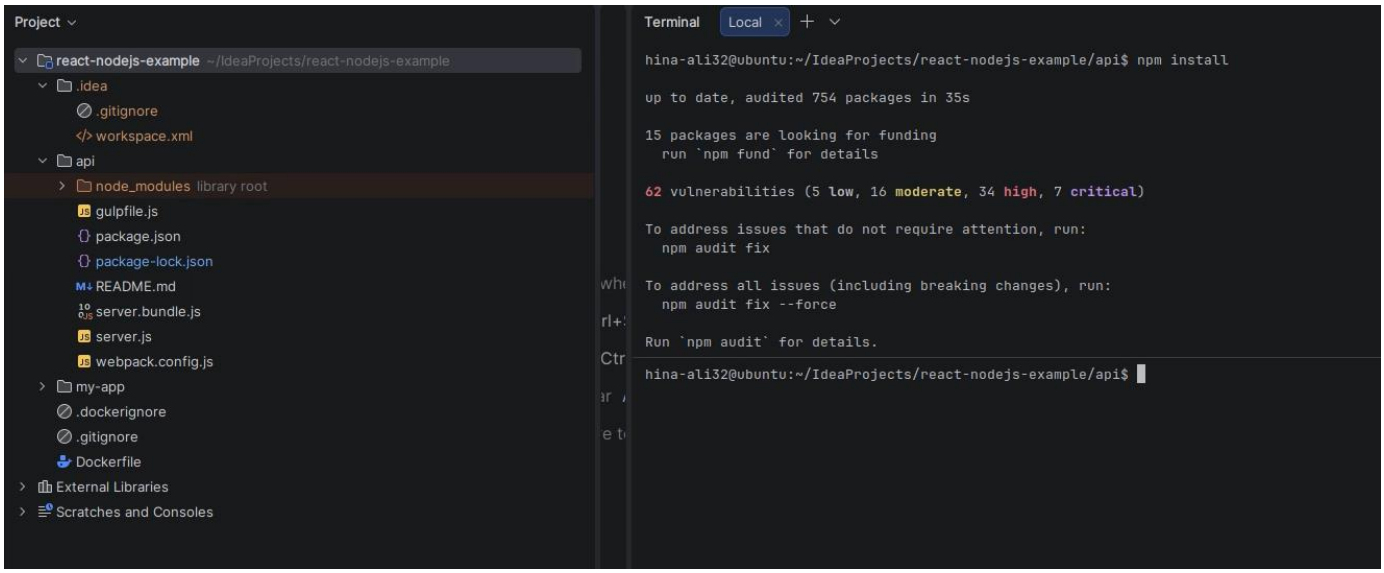
hina-ali32@ubuntu:~/IdeaProjects/node-project/app$
```

Shows npm pack creating bootcamp-node-project-1.0.0.tgz. The name and version come from package.json, and the tarball packages the application files for distribution.

Task 5: Build the React and Node.js API Artifact



Shows the Clone Repository dialog with the react-nodejs-example URL and destination directory. The repository is cloned from GitHub into /config/IdeaProjects/react-nodejs-example.



Shows npm install completing in the api directory and the api/node_modules folder in the Project pane. The node_modules folder holds the installed dependencies of the API.

```
hina-ali32@ubuntu:~/IdeaProjects/react-nodejs-example/api$ npm run build

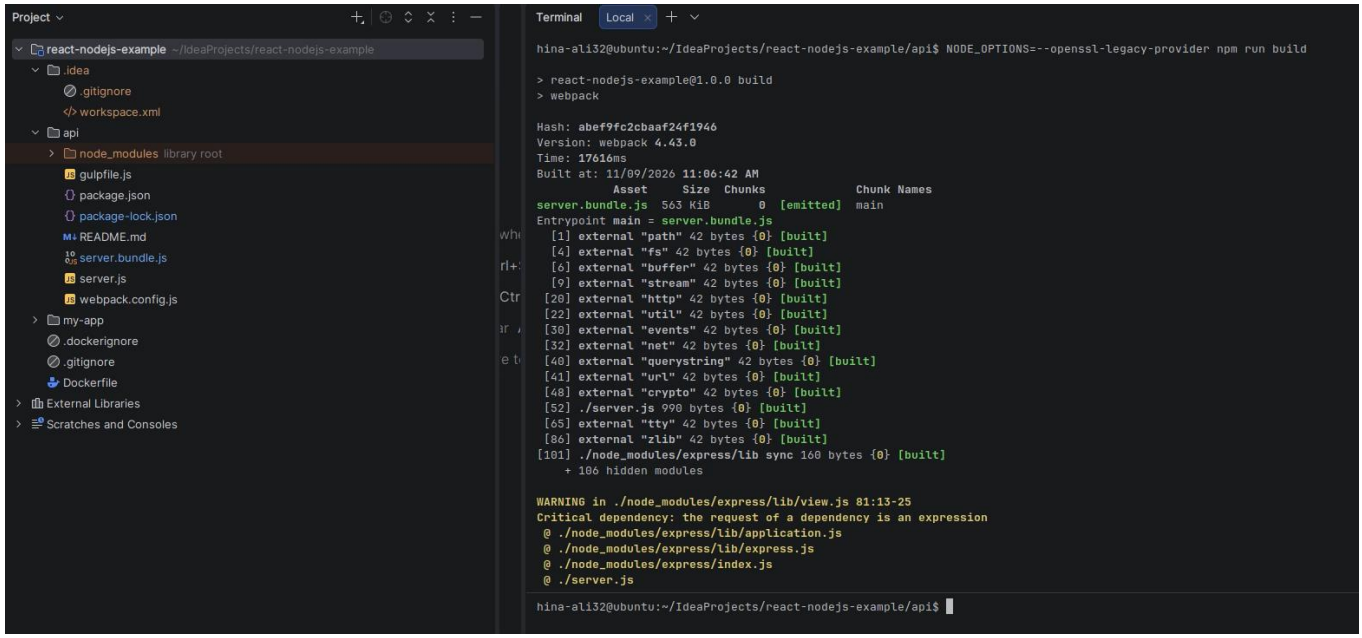
> react-nodejs-example@1.0.0 build
> webpack

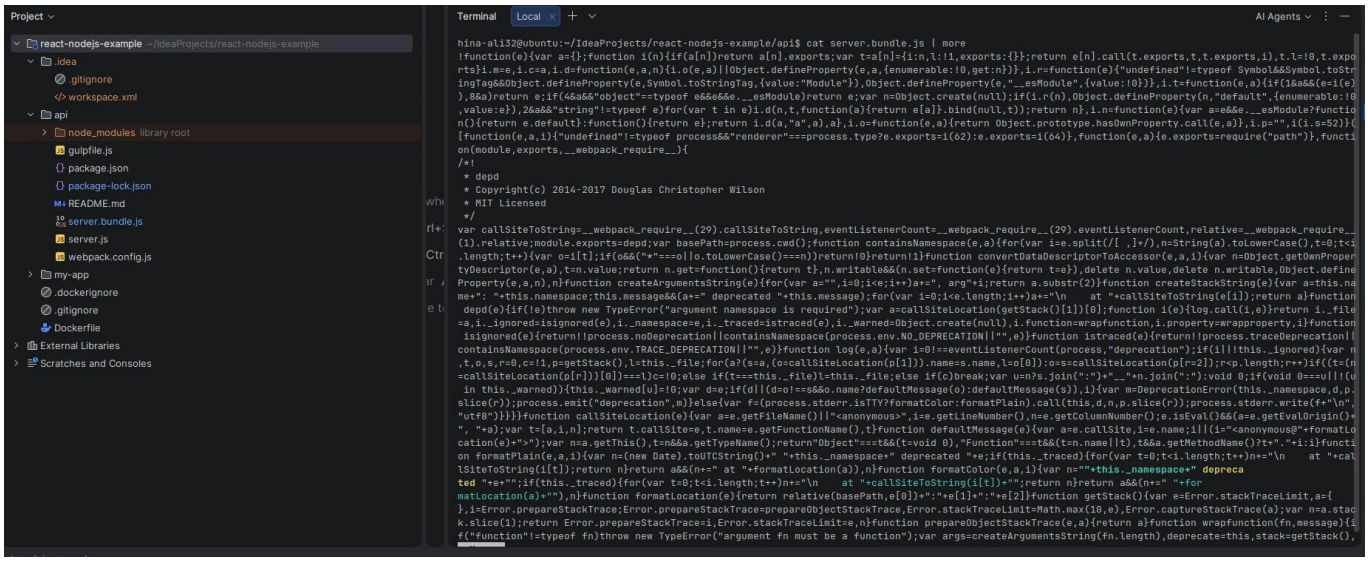
node:internal/crypto/hash:101
  this[kHandle] = new _Hash(algorithm, xofLen, algorithmId, getHashCache());
                  ^

Error: error:0308010C:digital envelope routines::unsupported
    at new Hash (node:internal/crypto/hash:101:19)
    at Object.createHash (node:crypto:139:10)
    at module.exports (/config/IdeaProjects/react-nodejs-example/api/node_modules/webpack/lib/util/createHash.js:135:53)
    at NormalModule._initBuildHash (/config/IdeaProjects/react-nodejs-example/api/node_modules/webpack/lib/NormalModule.js:417:16)
    at handleParseError (/config/IdeaProjects/react-nodejs-example/api/node_modules/webpack/lib/NormalModule.js:471:10)
    at /config/IdeaProjects/react-nodejs-example/api/node_modules/webpack/lib/NormalModule.js:503:5
    at /config/IdeaProjects/react-nodejs-example/api/node_modules/webpack/lib/NormalModule.js:358:12
    at /config/IdeaProjects/react-nodejs-example/api/node_modules/loader-runner/lib/LoaderRunner.js:373:3
    at iterateNormalLoaders (/config/IdeaProjects/react-nodejs-example/api/node_modules/loader-runner/lib/LoaderRunner.js:214:10)
    at Array.<anonymous> (/config/IdeaProjects/react-nodejs-example/api/node_modules/loader-runner/lib/LoaderRunner.js:205:4)
    at Storage.finished (/config/IdeaProjects/react-nodejs-example/api/node_modules/enhanced-resolve/lib/CachedInputFileSystem.js:55:16)
    at /config/IdeaProjects/react-nodejs-example/api/node_modules/enhanced-resolve/lib/CachedInputFileSystem.js:91:9
    at /config/IdeaProjects/react-nodejs-example/api/node_modules/graceful-fs/graceful-fs.js:123:16
    at FSReqCallback.readFileAfterClose [as oncomplete] (node:internal/fs/read/context:68:3) {
  opensslErrorStack: [
    'error:03000086:digital envelope routines::initialization error',
    'error:0308010C:digital envelope routines::unsupported'
  ],
  library: 'digital envelope routines',
  reason: 'unsupported',
  code: 'ERR_OSSL_EVP_UNSUPPORTED'
}

Node.js v22.23.4
hina-ali32@ubuntu:~/IdeaProjects/react-nodejs-example/api$
```

Shows the successful webpack build and server.bundle.js in the api directory. The build ran with NODE_OPTIONS=--openssl-legacy-provider, because this older webpack version (4.43.0) fails with ERR_OSSL_EVP_UNSUPPORTED on the newer Node.js.





Shows the bundled server.bundle.js displayed with cat server.bundle.js | more. The pager shows the first page of the bundled JavaScript, and the terminal prompt remains visible.

Task 6: Build a Docker Image for the Node.js Project



Shows the installed Docker version in IntelliJ's terminal. docker --version confirms that the Docker command is available through the host's Docker Engine.



Shows the node-project root directory containing the Dockerfile. The ls output lists the Dockerfile next to the app folder.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project$ sudo docker images

IMAGE                                  ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest   4f3b5ef873d2       11.6GB         3.01GB    U
mongo:8.0                              4a4ab5d48342       1.28GB          319MB    U
mysql:8.4                              76f0358884cb       1.12GB          254MB    U

hina-ali32@ubuntu:~/IdeaProjects/node-project$
```

Shows the existing Docker images before the build. This list serves as a baseline for comparison with the list after building.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project$ sudo docker build -t lab5:1.0 .
[+] Building 357.9s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 156B
 => [internal] load metadata for docker.io/library/node:13-alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 37.92MB
 => [1/5] FROM docker.io/library/node:13-alpine@sha256:527c70f74817f6f6b5853588c28de33459178ab72421f1fb7b63a281ab670258
 => => resolve docker.io/library/node:13-alpine@sha256:527c70f74817f6f6b5853588c28de33459178ab72421f1fb7b63a281ab670258
 => => sha256:780514bed1adda642a7f7008ea5b1477bb78a23e3f76eabbdde21031b2185a2f 35.30MB / 35.30MB
 => => sha256:4b9536424fa1675675e7edd59ae9a92573f0c010c5458733a493e2c3e1daf1ae 283B / 283B
 => => sha256:5d74fb112a7d313344f9f679ccb975062377464235fac36e1b952aa2b125d5e2 2.24MB / 2.24MB
 => => sha256:cbdbe7a5bc2a134ca8ec91be58565ec07d037386d1f1d8385412d224deafca08 2.81MB / 2.81MB
 => => extracting sha256:cbdbe7a5bc2a134ca8ec91be58565ec07d037386d1f1d8385412d224deafca08
 => => extracting sha256:780514bed1adda642a7f7008ea5b1477bb78a23e3f76eabbdde21031b2185a2f
 => => extracting sha256:5d74fb112a7d313344f9f679ccb975062377464235fac36e1b952aa2b125d5e2
 => => extracting sha256:4b9536424fa1675675e7edd59ae9a92573f0c010c5458733a493e2c3e1daf1ae
 => [2/5] COPY app/ /usr/app/
 => [3/5] WORKDIR /usr/app
 => [4/5] RUN npm install
 => [5/5] RUN npm run test
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:f8b1f3f269cb9cf08a2389c5bbaab915c7e77087ef5c4927e76776e6dfb1bef5
 => => exporting config sha256:f561ae4d80f42f01e31b104d7692a6b0cb34bea41d3d86ce7d4293b7378f3394
 => => exporting attestation manifest sha256:b92f5f70d32f4666bb9a0dd9a3be2869dcc5d7454b4c53399fd3e8ae8f2a76c4
 => => exporting manifest list sha256:7de2f7ea79c3b60de5b55e79f69bab03ed0d125791e1ffda3f3e09832eb6cb53
 => => naming to docker.io/library/lab5:1.0
 => => unpacking to docker.io/library/lab5:1.0

hina-ali32@ubuntu:~/IdeaProjects/node-project$
```

Shows sudo docker build -t lab5:1.0 . completing and tagging the image. Docker used the Dockerfile in the current directory, copied the app folder, installed the dependencies and ran the npm test during the build.

```
hina-ali32@ubuntu:~/IdeaProjects/node-project$ sudo docker images

IMAGE                                  ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest   4f3b5ef873d2       11.6GB         3.01GB   U
lab5:1.0                               7de2f7ea79c3        417MB         85.2MB
mongo:8.0                              4a4ab5d48342       1.28GB          319MB   U
mysql:8.4                              76f0358884cb       1.12GB          254MB   U

hina-ali32@ubuntu:~/IdeaProjects/node-project$
```

Shows the lab5 repository with tag 1.0 and its image ID in the image list. Together, the repository and tag identify the new image lab5:1.0.

Exam Evaluation Questions

1. **Maven built java-maven-app-1.1.0-SNAPSHOT.jar. Which values in pom.xml determine this name, and what does the package phase do?**

- The JAR name comes from pom.xml: artifactId (java-maven-app) + version (1.1.0-SNAPSHOT) = java-maven-app-1.1.0-SNAPSHOT.jar.
- The package phase runs the earlier steps (compile, test) and then packs the compiled code into a JAR in the target folder.
- Spring Boot then repackages it into an executable JAR, and the original is kept as .jar.original.

2. **Why can ./gradlew build work without a separate system-wide Gradle installation? Which JDK settings would you check if this sample reports an unsupported Java version?**

- ./gradlew build works because the project includes the Gradle Wrapper (gradlew and its properties file). On the first run, it downloads the required Gradle version (7.0) automatically, so no system-wide Gradle is needed.
- If an unsupported Java version error appears, these settings should be checked:
    1. Project SDK (File → Project Structure) should be JDK 16.
    2. Gradle JVM (Settings → Build Tools → Gradle) should be JDK 16.
    3. ./gradlew --version should show JVM 16 in the terminal.
- Gradle 7.0 cannot run on Java 21, so JDK 16 is required.

3. **Why does EADDRINUSE occur on port 3000 in this environment, and how does changing the application listener to port 4000 let the container's browser open it?**

- EADDRINUSE means port 3000 is already in use. The desktop (Webtop) uses it in this container, and only one program can use a port at a time.
- Changing the listener to port 4000 uses a free port, so the application starts without conflict.
- The browser runs inside the same container, so http://localhost:4000 reaches the application directly.